

本次课内容

海森堡方程（接上次课）

§ 4. A 在外磁场中的原子

复习1

海森堡方程（接上次课）

回顾：从薛定谔方程到海森堡方程

$$\bar{F} = \langle \psi(t) | F | \psi(t) \rangle \quad (\text{薛定谔绘景})$$

$$= \langle U \psi(0) | F | U \psi(0) \rangle$$

$$= \langle \psi(0) | \mathbf{U}^+ F \mathbf{U} | \psi(0) \rangle$$

$$\begin{aligned} i\hbar \dot{|\psi\rangle} &= H|\psi\rangle \\ \rightarrow |\psi\rangle &= U|\psi(0)\rangle \\ U &= e^{-\frac{i}{\hbar} H t} \end{aligned}$$

定义：

$$|\psi_H\rangle = |\psi(0)\rangle \quad (\text{不含 } t)$$

$$= U^+ |\psi(t)\rangle$$

$$F_H = U(t)^+ F U(t)$$

$$\bar{F} = \langle \psi_H | F_H(t) | \psi_H \rangle \quad (\text{海森堡绘景})$$

海森方程 (海森堡绘景)

$$F_H = U^\dagger(t) F U(t) \quad (U = e^{-\frac{i}{\hbar} H t}; U^\dagger = e^{+\frac{i}{\hbar} H t}),$$

$$\rightarrow \text{运动方程为 } \frac{dF_H}{dt} = \frac{1}{i\hbar} [F_H, H]; \quad \frac{d|\psi\rangle_H}{dt} = 0$$

(推导留作练习 (后附答案))

答案

$$F_H = U^\dagger F U$$



$$\frac{d}{dt} F_H(t) = U^\dagger F \left(\frac{d}{dt} U \right) + \left(\frac{d}{dt} U^\dagger \right) F U$$

$$= U^\dagger F \left(-\frac{i}{\hbar} H U + \frac{i}{\hbar} U^\dagger H F U \right)$$

$$= -\frac{i}{\hbar} \left(\underline{U^\dagger F U H} - H \underline{U^\dagger F U} \right)$$

$$U(t) = e^{-\frac{i}{\hbar} H t}$$

$$U^\dagger(t) = e^{+\frac{i}{\hbar} H t}$$



$$\frac{d}{dt} U = -\frac{i}{\hbar} H U$$

$$\frac{d}{dt} U^\dagger = +\frac{i}{\hbar} U^\dagger H$$

$$(U^\dagger H = H U^\dagger)$$

$$\frac{dF_H}{dt} = \frac{1}{i\hbar} [F_H, H]$$

(海森堡方程)

例题：写出一维谐振子的海森堡方程

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

$$[x, p] = i\hbar$$

$$\begin{aligned}\dot{x} &= \frac{1}{i\hbar} [x, H] = \frac{[x, p^2]}{i\hbar 2m} = \frac{[x, p]p + p[x, p]}{i\hbar 2m} \\ &= \frac{p}{m}\end{aligned}$$

(一维谐振子也可以用升降算符方法)

$$\dot{p} = \frac{1}{i\hbar} [p, H] = \frac{1}{i\hbar} [p, \frac{1}{2}m\omega^2 x^2] = -m\omega^2 x$$

$$\left([p, U(x)] = -i\hbar \frac{\partial}{\partial x} U \right)$$

*作业补充题 4.5.2 (选做)

证明谐振子相干态波包宽度不随时间变化

- 提示：可以求解海森堡方程

$$\frac{d}{dt}a = \frac{1}{i\hbar}[a, H], \frac{d}{dt}a^+ = \frac{1}{i\hbar}[a^+, H], \text{ 其中: } H = \hbar\omega(a^+a + \frac{1}{2})$$

$$\Rightarrow \frac{d}{dt}a = -i\omega a, \frac{d}{dt}a^+ = i\omega a^+$$

$$\Rightarrow a(t) = a(0)e^{-i\omega t}, \quad a^+(t) = a^+(0)e^{i\omega t}$$

相干态 $|\beta\rangle$ 的定义: $a(0)|\beta\rangle = \beta|\beta\rangle$

$$x'(t) = \frac{1}{\sqrt{2}}[a(t) + a^+(t)], \quad \overline{(\Delta x')^2} = \overline{(x' - \overline{x'})^2}, \quad \text{波包宽度: } \sqrt{\overline{(\Delta x')^2}}$$

• 例1：由海森堡方程给出守恒量满足的条件
(假设算符力学量算符F在薛定谔绘景中不含t)

解法一 $\frac{dF_H}{dt} = \frac{1}{i\hbar} [F_H, H] \Rightarrow$

若 $[F, H] = 0$, 则 $[F, U] = 0$,

$$U(t) = e^{-\frac{i}{\hbar} H t}$$

$$F_H = U^\dagger F U = U^\dagger U F = F$$

$$\rightarrow \frac{dF_H}{dt} = \frac{1}{i\hbar} [F_H, H] = \frac{1}{i\hbar} [F, H] = 0, \text{ 所以}$$

$$\frac{d\bar{F}}{dt} = \langle \psi_H | \frac{dF_H(t)}{dt} | \psi_H \rangle = 0$$

(对任意态成立, 所以 F 是守恒量)

解法二

$$U(t) = e^{-\frac{i}{\hbar} H t}$$

由海森堡方程的解 $F_H(t) = U^\dagger(t) F U(t)$

若满足 $[F, H] = 0 \Rightarrow [F, U] = 0$, 则

$F_H(t) = U^\dagger F U = U^\dagger U F = F$ 不含时间, 所以

$\bar{F} = \langle \psi_H | F_H(t) | \psi_H \rangle = \langle \psi_H | F | \psi_H \rangle$ 不含 t 。

(对任意态成立, 所以 F 是守恒量)

例2

- 写出能量H表象下海森堡方程的矩阵形式，并求解。

解：

$$\frac{dF_H}{dt} = \frac{1}{i\hbar} [F_H, H] \Rightarrow$$
$$\frac{d(\hat{F}_H)_{mn}}{dt} = \left(\frac{d\hat{F}_H}{dt} \right)_{mn} = \frac{1}{i\hbar} \langle \mathbf{m} | (\hat{F}_H \hat{H} - \hat{H} \hat{F}_H) | \mathbf{n} \rangle$$
$$= \frac{1}{i\hbar} (E_n - E_m) \langle \mathbf{m} | \hat{F}_H | \mathbf{n} \rangle = \frac{1}{i\hbar} (E_n - E_m) (\hat{F}_H)_{mn} \Rightarrow$$

$$(\hat{F}_H)_{mn}(t) = (\hat{F}_H)_{mn}(0) e^{i\omega_{mn}t}$$

$$= (\hat{F})_{mn} e^{i\omega_{mn}t}; \text{ 其中: } \omega_{mn} = \frac{1}{\hbar} (E_m - E_n)$$

矩阵力学

从薛定谔图景变换到海森堡图景，可以理解成力学量矩阵元乘了一个时间因子。

这一结果也可以由下面方法直接得到：

$$\mathbf{F}_H = \mathbf{U}^\dagger \mathbf{F} \mathbf{U} \Rightarrow (\mathbf{F}_H)_{mn} = (\mathbf{U}^\dagger \mathbf{F} \mathbf{U})_{mn}$$

$$= \langle m | e^{+\frac{i}{\hbar} H t} \mathbf{F} e^{-\frac{i}{\hbar} H t} | n \rangle = (\mathbf{F})_{mn} e^{i \omega_{mn} t}$$

矩阵力学

*经典力学的泊松括号形式:

正则方程 $\dot{q} = \frac{\partial H}{\partial p}; \quad \dot{p} = -\frac{\partial H}{\partial q};$

可以改写成:

$$\begin{aligned}\dot{q} &= \{q, H\}; \\ \dot{p} &= \{p, H\}\end{aligned}$$

泊松括号定义为

$$\{A, B\} \equiv \frac{\partial A}{\partial q} \frac{\partial B}{\partial p} - \frac{\partial A}{\partial p} \frac{\partial B}{\partial q}$$

从经典到量子的过渡

经典 (数)

$$\dot{q} = \{q, H\};$$
$$\dot{p} = \{p, H\}$$

量子 (算符)

$$\dot{q} = \frac{1}{i\hbar} [q, H];$$
$$\dot{p} = \frac{1}{i\hbar} [p, H]$$

量子化条件

$$[q, p] = i\hbar$$

(不对易代数)

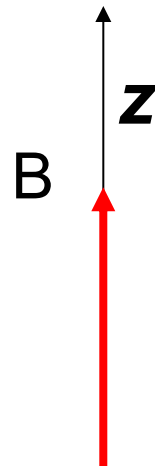
(海森堡方程)

(等价于薛定谔方程)

原子中电子
的轨道磁矩

$$\vec{M}_L = -\frac{e}{2m_e} \vec{L}$$

(e 是电子电量的绝对值)



电子自旋磁矩: $\vec{M}_S = -\frac{e}{m_e} \vec{S}$

忽略原子核自旋磁矩和核绕质心运动的轨道磁矩
($m_e \ll m_N$)

原子总磁矩: $\vec{M} = \vec{M}_L + \vec{M}_S = -\frac{e}{2m_e} (\vec{L} + 2\vec{S})$

回顾：常用力学量完全集：

1 x, y, z, S_z

2 p_x, p_y, p_z, S_z

3 孤立氢原子： H, L^2, L_z, S_z

$$H_0 = -\frac{\hbar^2}{2\mu} \nabla^2 + U(r) \quad (\text{无磁场时})$$

磁势能 $U_m = -\vec{M} \cdot \vec{B} = -M_z B = \frac{eB}{2m_e} (\hat{L}_z + 2\hat{S}_z).$

$$H = H_0 + \frac{eB}{2m_e} (\hat{L}_z + 2\hat{S}_z)$$

$H_0, L^2, \hat{L}_z, \hat{S}_z$ 彼此对易，取为力学量完全集，
取其共同本征函数：

$$R_{nl}(r)Y_{lm_l}(\theta, \varphi) \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ 和 } R_{nl}(r)Y_{lm_l}(\theta, \varphi) \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

H_0 本征值记为 $E_n^{(0)}$ （显然也是 H 的本征函数）

$$H = H_0 + \frac{eB}{2m_e} (\hat{L}_z + 2\hat{S}_z)$$

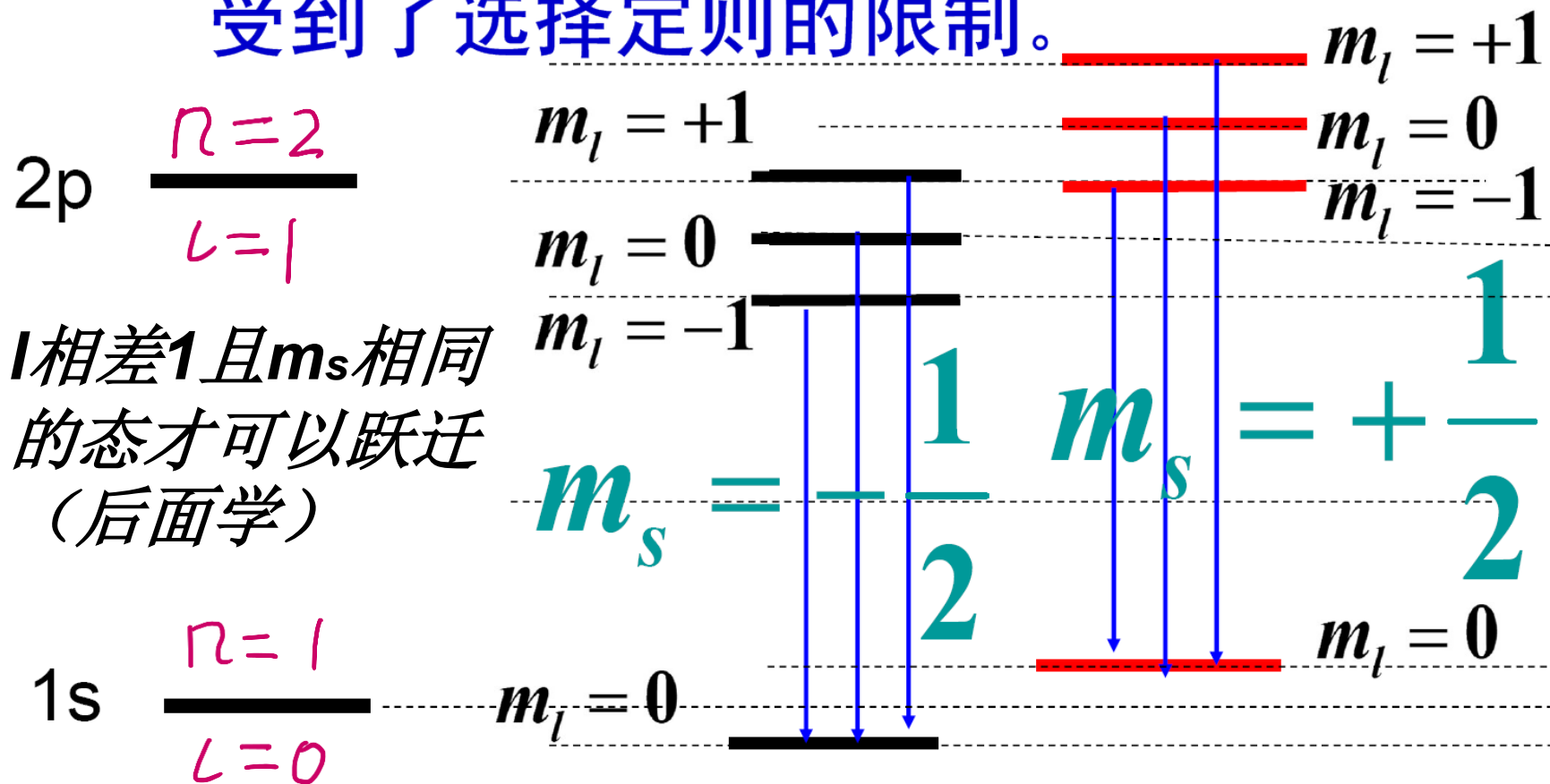
$$\mathbf{E} = E_n^{(0)} + \frac{eB\hbar}{2m_e} (m_l + 2m_s)$$

$$(m_s = \pm \frac{1}{2})$$

*若改为碱金属原子，
则 $E_n^{(0)}$ 改为 $E_{nl}^{(0)}$

$$\psi_{n,l,m_l,m_s} = \begin{cases} \psi_{n,l,m_l,1/2} = R_{nl}(r)Y_{lm_l}(\theta,\varphi) \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ \psi_{n,l,m_l,-1/2} = R_{nl}(r)Y_{lm_l}(\theta,\varphi) \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{cases}$$

* 实验上只有三条谱线，这是由于受到了选择定则的限制。



$$E_{n l m_l m_s} = E_n^{(0)} + \frac{eB\hbar}{2m_e} \cdot (m_l + 2m_s) \quad (m_s = \pm \frac{1}{2})$$

*利用 **Zeeman** 效应可以

1. 调节原子能级
2. 操控原子的量子态（结合激光）
——磁光阱（**MOT**）、相干控制、量子态操控等。
3. 灵敏检测磁场（结合光谱技术）
——原子磁强计。

复习1

例题讲解

自旋矩阵 (S_z 表象)

$$S_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, S_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

没有共同本征态!

记住!

$\vec{S}$ 最多只有一个分量具有确定的取值

$$\vec{M}_S = -\frac{e}{m} \vec{S} \quad (\text{e是电子电量的绝对值})$$

容易验证：自旋矩阵满足

$$[S_x, S_y] = i\hbar S_z, \quad [S_y, S_z] = i\hbar S_x, \quad [S_z, S_x] = i\hbar S_y,$$

$$S_x^2 = S_y^2 = S_z^2 = \frac{\hbar^2}{4}$$

电子自旋角动量: $\hat{\vec{S}} = \frac{\hbar}{2} \vec{\sigma}$.

引入Pauli矩阵 (无量纲化)

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$


$$\hat{\vec{S}} = \frac{\hbar}{2} \vec{\sigma}.$$

$$\sigma_x \sigma_y = -\sigma_y \sigma_x = i\sigma_z, \rightarrow [\sigma_x, \sigma_y] = 2i\sigma_z$$

$$x \rightarrow y \rightarrow z \rightarrow x \quad \sigma_x^2 = \sigma_y^2 = \sigma_z^2 = I,$$

例1 利用自旋矩阵证明自旋角动量 $\mathbf{S}$ 在任意方向 $\mathbf{n}$ ($\mathbf{n}$ 为单位向量, 长度为1) 的投影 $\mathbf{S}_n$ 的测量值都是 $\pm \frac{\hbar}{2}$

证明: $\hat{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}; \hat{S}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}; \hat{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$$\mathbf{S}_n \equiv \vec{\mathbf{S}} \bullet \vec{\mathbf{n}} = S_x n_x + S_y n_y + S_z n_z$$

$$n_x^2 + n_y^2 + n_z^2 = 1$$

(要点)

$$S_n \equiv \vec{S} \cdot \vec{n} = S_x n_x + S_y n_y + S_z n_z$$

$$= \frac{\hbar}{2} \begin{bmatrix} n_z & n_x - i n_y \\ n_x + i n_y & -n_z \end{bmatrix}$$

$$n_x^2 + n_y^2 + n_z^2 = 1$$

由行列式

$$\begin{vmatrix} \frac{\hbar}{2} n_z - \lambda & \frac{\hbar}{2} (n_x - i n_y) \\ \frac{\hbar}{2} (n_x + i n_y) & -\frac{\hbar}{2} n_z - \lambda \end{vmatrix} = 0$$

得本征值 $\lambda = \pm \frac{\hbar}{2} \sqrt{n_x^2 + n_y^2 + n_z^2} = \pm \frac{\hbar}{2}$

例2.

- 在 S_z 本征态 $(+\frac{\hbar}{2})$ 下测量 S_n , S_n 平均值是多大?

$$\langle S_n \rangle = (1, 0) S_n \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= (1, 0) \frac{\hbar}{2} \begin{bmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$


$$= (1, 0) \frac{\hbar}{2} \begin{bmatrix} n_z \\ n_x + in_y \end{bmatrix} = \frac{\hbar}{2} n_z$$

例3

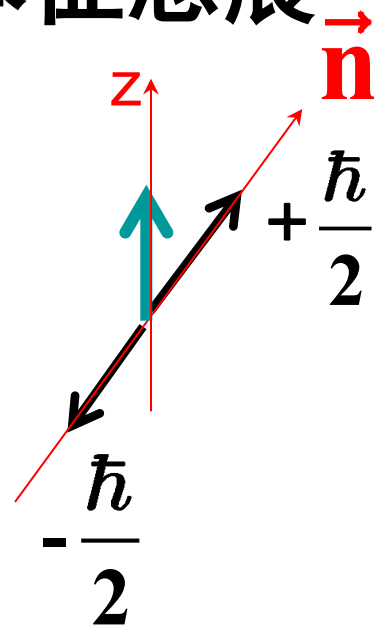
- 在 S_z 本征态 $(+\frac{\hbar}{2})$ 下测量 S_n , 结果如何? 概率分布是怎样的?

分析 1. 先求 S_n 本征态 (本征向量)

2. 将 S_z 本征态按照 S_n 本征态展开. 系数的模平方即所求几率.

$$S_z \text{ 本征态是 } |\psi\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

S_n 本征态是什么?



解法一：直角坐标法

$$n_x^2 + n_y^2 + n_z^2 = 1$$

S_n 本征值 $+\frac{\hbar}{2}$ 对应的本征方程为

$$\frac{\hbar}{2} \begin{bmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = +\frac{\hbar}{2} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

$$n_z c_1 + (n_x - in_y) c_2 = c_1$$

$$c_2 = \frac{(1 - n_z)}{n_x - in_y} c_1 \quad (1)$$

由归一化, $|\mathbf{c}_1|^2 + |\mathbf{c}_2|^2 = 1$ 得

$$|\mathbf{c}_1|^2 + \left| \frac{1-n_z}{n_x - in_y} \right|^2 |\mathbf{c}_1|^2 = 1, \quad |\mathbf{c}_1|^2 + \frac{(1-n_z)^2}{n_x^2 + n_y^2} |\mathbf{c}_1|^2 = 1,$$

$$|\mathbf{c}_1|^2 + \frac{(1-n_z)^2}{1-n_z^2} |\mathbf{c}_1|^2 = 1, \quad |\mathbf{c}_1|^2 + \frac{1-n_z}{1+n_z} |\mathbf{c}_1|^2 = 1,$$

$$\text{取 } \mathbf{c}_1 = \sqrt{\frac{1+n_z}{2}} \quad (2)$$

(2) 代入 (1) 得 $\mathbf{c}_2$

$$\begin{aligned}
 c_2 &= \frac{(1-n_z)}{n_x - in_y} c_1 = \frac{(1-n_z)}{n_x - in_y} \sqrt{\frac{1+n_z}{2}} \\
 &= \frac{(1-n_z)(n_x + in_y)}{n_x^2 + n_y^2} \sqrt{\frac{1+n_z}{2}} \\
 &= \frac{(1-n_z)}{1-n_z^2} \sqrt{\frac{1+n_z}{2}} (n_x + in_y) = \frac{n_x + in_y}{\sqrt{2}(1+n_z)} \rightarrow |n_+\rangle = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} \sqrt{\frac{1+n_z}{2}} \\ \frac{n_x + in_y}{\sqrt{2}(1+n_z)} \end{bmatrix}
 \end{aligned}$$

同理，另一个本征值对应的本征方程为

$$\frac{\hbar}{2} \begin{bmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = -\frac{\hbar}{2} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

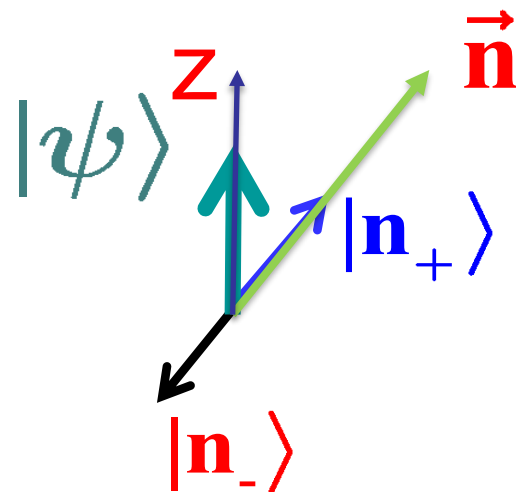
用同样的解法得到 $|n_-\rangle = \begin{bmatrix} \sqrt{\frac{1-n_z}{2}} \\ -\frac{n_x + in_y}{\sqrt{2}(1-n_z)} \end{bmatrix}$

$$|n_+\rangle = \begin{pmatrix} \sqrt{\frac{1+n_z}{2}} \\ \frac{n_x + in_y}{\sqrt{2(1+n_z)}} \end{pmatrix} \quad |n_-\rangle = \begin{pmatrix} \sqrt{\frac{1-n_z}{2}} \\ -\frac{n_x + in_y}{\sqrt{2(1-n_z)}} \end{pmatrix}$$

$$|\psi\rangle = c_+ |n_+\rangle + c_- |n_-\rangle$$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad c_+ = \langle n_+ | \psi \rangle = \sqrt{\frac{1+n_z}{2}}$$

$$c_- = \langle n_- | \psi \rangle = \sqrt{\frac{1-n_z}{2}}$$



S_n 测量值为 $\hbar / 2$ 的几率： $|c_+|^2 = (1 + n_z) / 2$

S_n 测量值为 $-\hbar / 2$ 的几率： $|c_-|^2 = (1 - n_z) / 2$

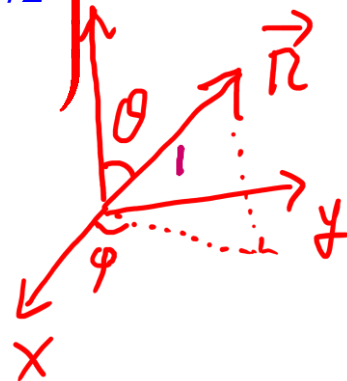
解法二：球坐标法

$$\vec{n}: \quad \underline{n_z = \cos \theta, \quad n_x = \sin \theta \cos \varphi, \quad n_y = \sin \theta \sin \varphi}$$

$$S_n = \frac{\hbar}{2} \begin{bmatrix} n_z & n_x - i n_y \\ n_x + i n_y & -n_z \end{bmatrix} = \frac{\hbar}{2} \begin{bmatrix} \cos \theta & \sin \theta e^{-i\varphi} \\ \sin \theta e^{i\varphi} & -\cos \theta \end{bmatrix}$$

$$\text{取 } |\mathbf{n}_+\rangle = \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\varphi/2} \\ \sin \frac{\theta}{2} e^{i\varphi/2} \end{pmatrix}; \quad |\mathbf{n}_-\rangle = \begin{pmatrix} \sin \frac{\theta}{2} e^{-i\varphi/2} \\ -\cos \frac{\theta}{2} e^{i\varphi/2} \end{pmatrix}$$

(θ 、 φ 都是常数)



$$|\psi\rangle = c_+ |n_+\rangle + c_- |n_-\rangle$$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |n_+\rangle = \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\varphi/2} \\ \sin \frac{\theta}{2} e^{i\varphi/2} \end{pmatrix}; \quad |n_-\rangle = \begin{pmatrix} \sin \frac{\theta}{2} e^{-i\varphi/2} \\ -\cos \frac{\theta}{2} e^{i\varphi/2} \end{pmatrix}$$

$$c_+ = \langle n_+ | \psi \rangle = \left(\cos \frac{\theta}{2} e^{i\varphi/2}, \sin \frac{\theta}{2} e^{-i\varphi/2} \right) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \cos \frac{\theta}{2} e^{i\varphi/2}$$

$$c_- = \langle n_- | \psi \rangle = \left(\sin \frac{\theta}{2} e^{i\varphi/2}, -\cos \frac{\theta}{2} e^{-i\varphi/2} \right) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \sin \frac{\theta}{2} e^{i\varphi/2}$$

$$S_n \text{ 测量值为 } \hbar/2 \text{ 的几率: } |c_+|^2 = \cos^2 \frac{\theta}{2}$$

$$S_n \text{ 测量值为 } -\hbar/2 \text{ 的几率: } |c_-|^2 = \sin^2 \frac{\theta}{2}$$

解法三 [简单解法]: 由平均值求概率分布

$$S_n \equiv \vec{S} \bullet \vec{n} = S_x n_x + S_y n_y + S_z n_z = \frac{\hbar}{2} \begin{bmatrix} n_z & n_x - i n_y \\ n_x + i n_y & -n_z \end{bmatrix}$$

$$\left\{ \begin{aligned} \langle S_n \rangle &= (\mathbf{1}, \mathbf{0}) S_n \begin{bmatrix} \mathbf{1} \\ \mathbf{0} \end{bmatrix} = \frac{\hbar}{2} n_z \end{aligned} \right.$$

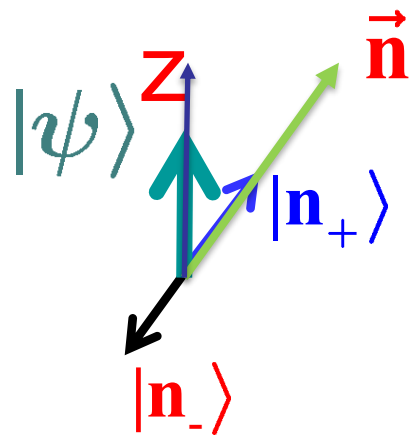
$$\langle S_n \rangle = \frac{\hbar}{2} |c_+|^2 + \left(-\frac{\hbar}{2}\right) |c_-|^2$$

$$\Rightarrow |c_+|^2 - |c_-|^2 = n_z$$

$$\text{归一化条件: } |c_+|^2 + |c_-|^2 = 1$$

$$S_n \text{ 测量值为 } \hbar/2 \text{ 的几率: } |c_+|^2 = (1 + n_z)/2$$

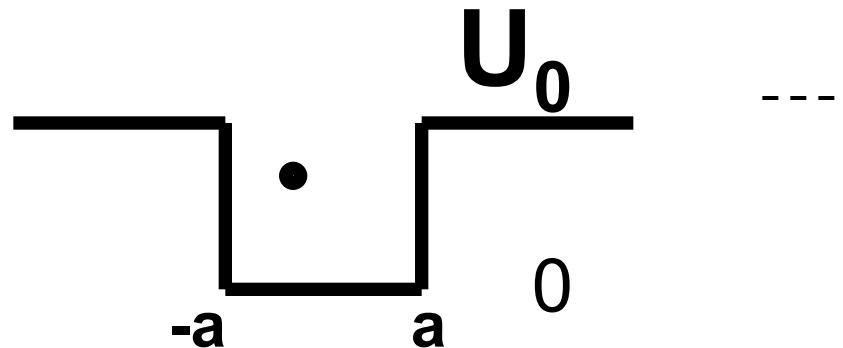
$$S_n \text{ 测量值为 } -\hbar/2 \text{ 的几率: } |c_-|^2 = (1 - n_z)/2$$



例4

证明有限深对称方势阱能量 $E > 0$

$$U(x) = \begin{cases} 0, & -a < x < a \\ U_0 (> 0), & x < -a \text{ or } x > a \end{cases}$$



$$H\psi = E\psi \Rightarrow$$

$$\int_{-\infty}^{\infty} \psi^* H \psi dx = E \int_{-\infty}^{\infty} \psi^* \psi dx = E \quad (1)$$

$$(\text{假设已归一化: } \int_{-\infty}^{\infty} \psi^* \psi dx = 1)$$

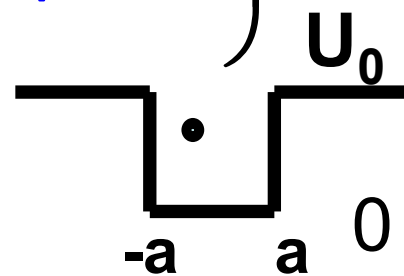
$$\text{将 } H = -\frac{\hbar^2}{2\mu} \frac{d^2}{dx^2} + U(x) \text{ 代入 (1) 左边,}$$

可以证明两项（动能和势能）的贡献都为正，
证明如下：

$$\text{其中} \int_{-\infty}^{\infty} \psi^* U(x) \psi dx = \int_{-\infty}^{\infty} U(x) |\psi|^2 dx > U_{\text{阱底}} = 0 \quad (2)$$

$$(\because U(x) \geq U_{\text{阱底}} = 0)$$

$$\begin{aligned} & \text{且} -\frac{\hbar^2}{2\mu} \int_{-\infty}^{\infty} \psi^* \frac{d^2}{dx^2} \psi dx \quad \left(\overline{\frac{p^2}{2\mu}} = \frac{\langle \psi | p^2 | \psi \rangle}{2\mu} = \frac{\langle p\psi | p\psi \rangle}{2\mu} \geq 0 \right) \quad \leftarrow U_0 (> 0) \\ &= -\frac{\hbar^2}{2\mu} \left(\psi^* \frac{d}{dx} \psi \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{d}{dx} \psi \frac{d}{dx} \psi^* dx \right) \\ &= \frac{\hbar^2}{2\mu} \int_{-\infty}^{\infty} \left| \frac{d}{dx} \psi \right|^2 dx \geq 0 \quad (3) \end{aligned}$$

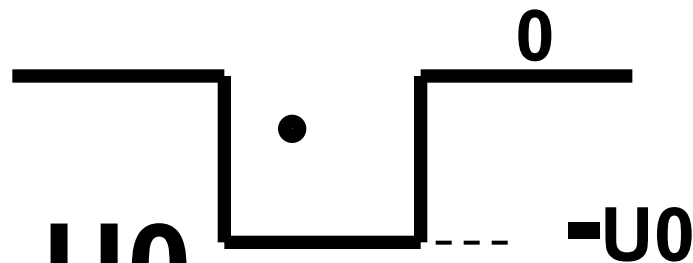


(以上用了分部积分, 并利用了 $\psi(\pm\infty) \rightarrow 0$)

将 (2) (3) 代入 (1) 得 $E > 0$ (得证)

思考题：如果势阱改为如下形式，

$$U(x) = \begin{cases} -U_0 (< 0), & -a < x < a \\ 0 & x < -a \text{ or } x > a \end{cases}$$



试证明 $E > -U_0$

提示：因 $U(x) \geq U_{\text{阱底}}$ ，故

$$\int_{-\infty}^{\infty} \psi^* U(x) \psi dx = \int_{-\infty}^{\infty} U(x) |\psi|^2 dx > U_{\text{阱底}} = -U_0,$$

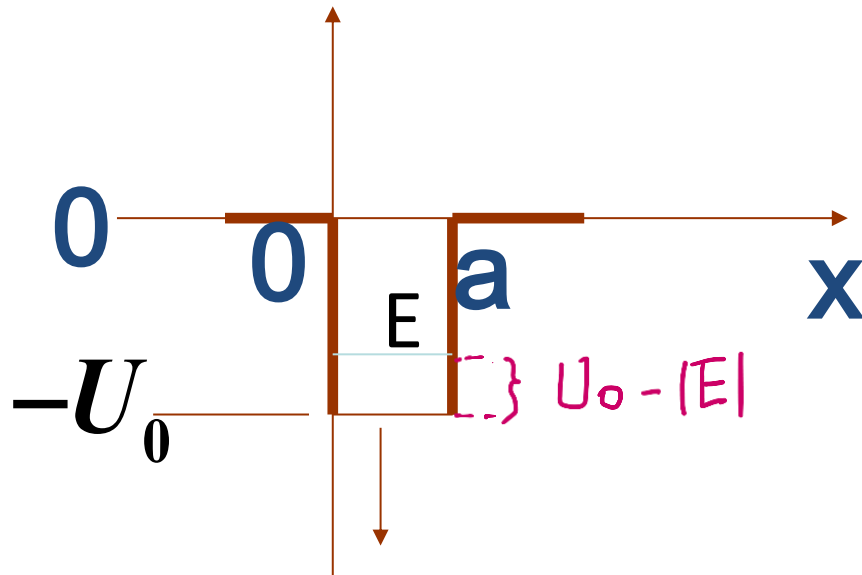
重复前面的证明可得 $E > -U_0$

结论：能量E总比势阱底部势能高。

例5

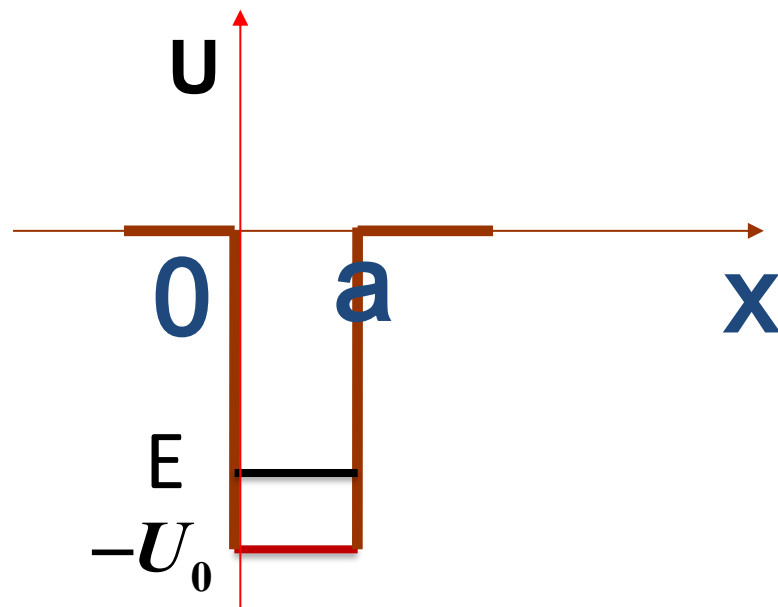
- 求解以下一维无限深势阱的束缚态能级和波函数
($-U_0 < E < 0$, $\underbrace{U_0 - |E| \ll U_0}$)

$$\left\{ \begin{array}{l} V(x) = 0 \quad (x < 0 \text{ 或 } x > a) \\ V(x) = -U_0 \quad (0 < x < a) \quad (U_0 \rightarrow +\infty) \end{array} \right.$$



解：这里只求束缚态解

阱内： $\frac{d^2\psi}{dx^2} + k^2\psi(x) = 0$



$$k = \sqrt{2\mu(E + U_0)} / \hbar \quad (-U_0 < E < 0, \quad U_0 - |E| \ll U_0)$$

(能量 E 接近阱底势能，远小于阱口势能)

$$\psi(x) = A \cos(kx) + B \sin(kx)$$

$$x = 0 \text{ 处 } \psi = 0 \Rightarrow A = 0, \quad \psi(x) = B \sin(kx)$$

$$x = a \text{ 处 } \psi = 0 \Rightarrow k = \frac{n\pi}{a} \quad (n = 1, 2, 3, \dots)$$

$$k = \sqrt{2\mu(E + U_0)} / \hbar = \frac{n\pi}{a} \Rightarrow$$

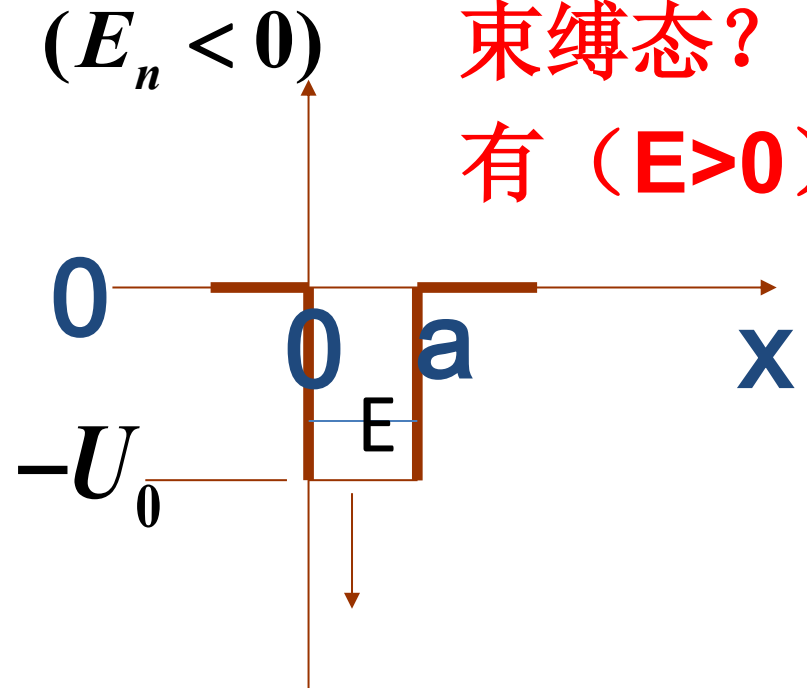
$$E_n = -U_0 + \frac{\hbar^2}{2\mu} \left(\frac{n\pi}{a} \right)^2$$

$$(n = 1, 2, 3, \dots)$$

最后归一化给出 B ,

$$\psi_n(x) = \begin{cases} \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right) & (0 \leq x \leq a) \\ 0 & (x < 0 \text{ or } x > a) \end{cases}$$

思考题：
有没有非
束缚态？
有 ($E > 0$)



例7：对一维无限深势阱中的粒子处于基态，求出其动量的几率分布，并讨论是否随时间变化。

解

$$\psi_n(x) = \begin{cases} \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) & 0 < x < a \\ 0 & x > a, x < 0 \end{cases}$$

($n=1, 2, 3, \dots$)

定态波函数： $\psi_n(x) \exp(-\frac{i}{\hbar} E_n t)$

$$\psi_1(x) = \begin{cases} \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right) & 0 < x < a \\ 0 & x > a, x < 0 \end{cases}$$

基态波函数： $\psi_1(x) \exp(-\frac{i}{\hbar} E_1 t)$

从而动量几率幅为

$$\begin{aligned} c(p, t) &= \int_{-\infty}^{\infty} \varphi_p^*(x) \psi_1(x) \exp(-\frac{i}{\hbar} E_1 t) dx \\ &= \int_0^a \frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i}{\hbar} px} \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right) dx \exp(-\frac{i}{\hbar} E_1 t) \\ &= \int_0^a \frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i}{\hbar} px} \sqrt{\frac{2}{a}} \frac{e^{i\frac{\pi x}{a}} - e^{-i\frac{\pi x}{a}}}{2i} dx \exp(-\frac{i}{\hbar} E_1 t) \end{aligned}$$

$$C_p \equiv c(p, t) = \sqrt{\frac{2a}{h}} \frac{1}{\pi} \frac{1}{\left(\frac{2pa}{h}\right)^2 - 1} \left(1 + e^{-i2\pi\left(\frac{pa}{h}\right)} \right) \exp\left(-\frac{i}{h} E_1 t\right)$$

$$|C_p|^2 = \frac{2a}{h} \frac{1}{\pi^2} \frac{1}{\left[\left(\frac{2pa}{h}\right)^2 - 1\right]^2} 4 \cos^2 \left[\pi \left(\frac{pa}{h} \right) \right]$$

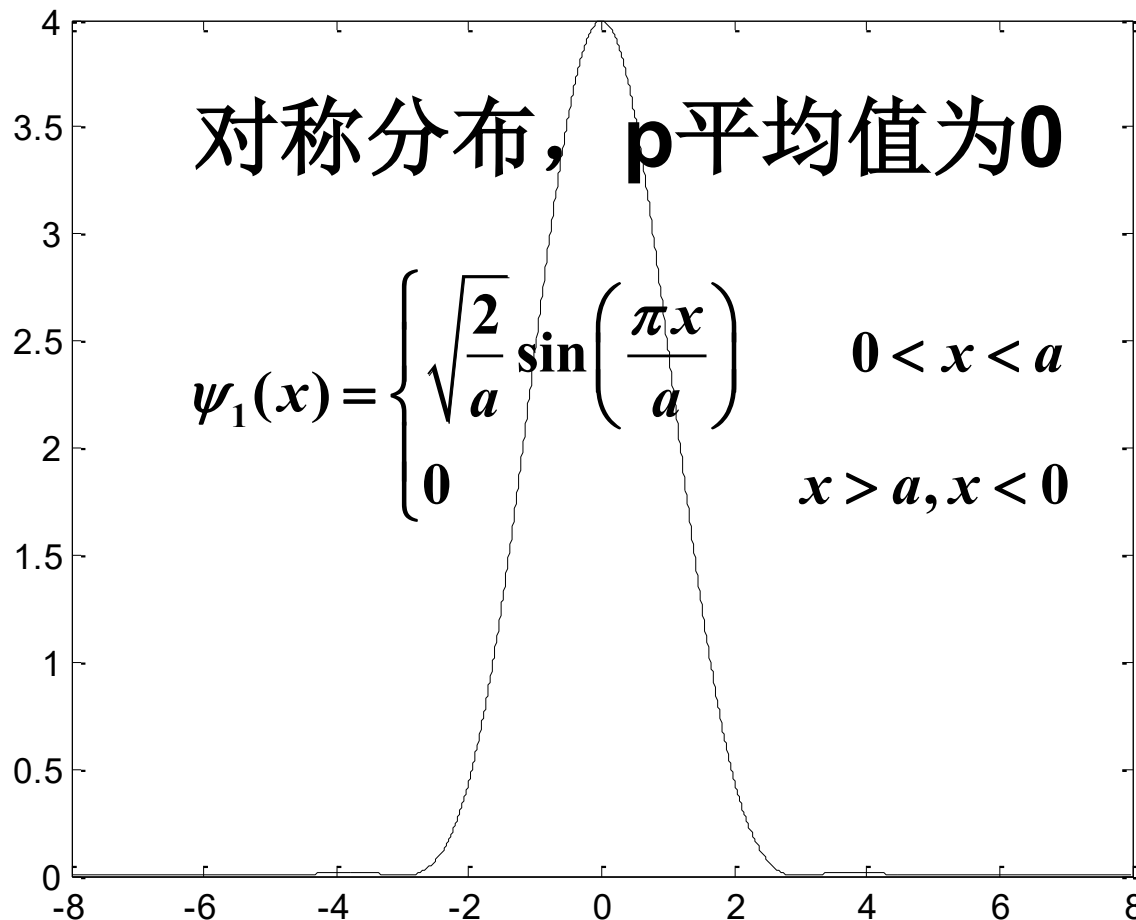
(思考：为什么不含t?)

是因为定态还是守恒量？

基态(n=1)动量分布

$$|c_p|^2 \frac{h}{2a}$$

几
率
密
度



无量纲动量

pa/h

$$\psi_n(x) = \begin{cases} \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) & 0 < x < a \\ 0 & x > a, x < 0 \end{cases}$$

($n=1, 2, 3, \dots$)

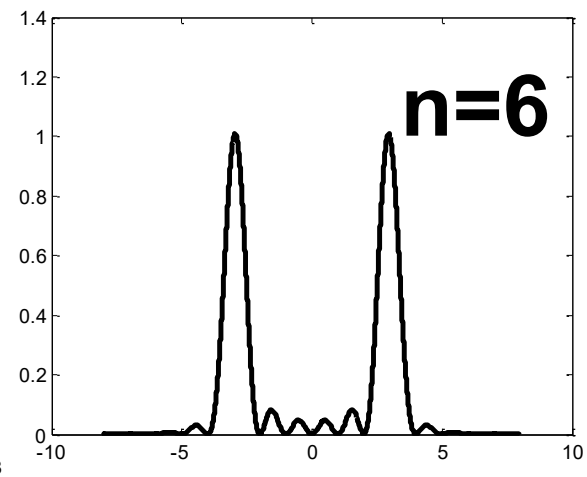
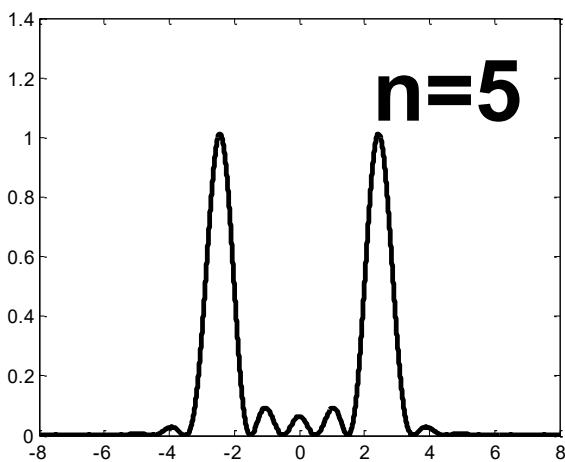
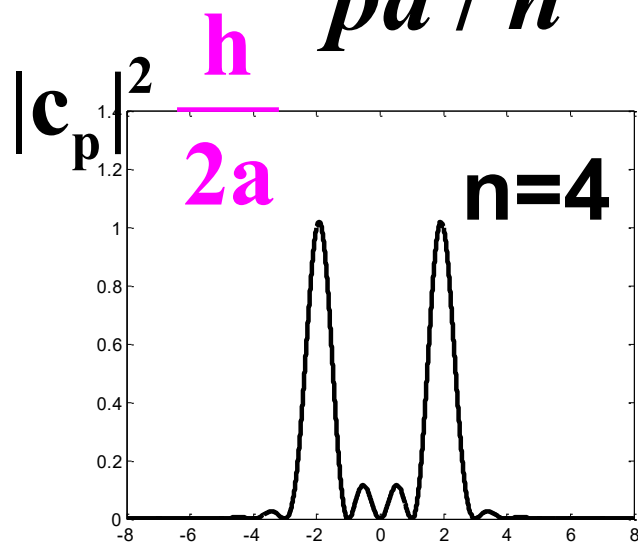
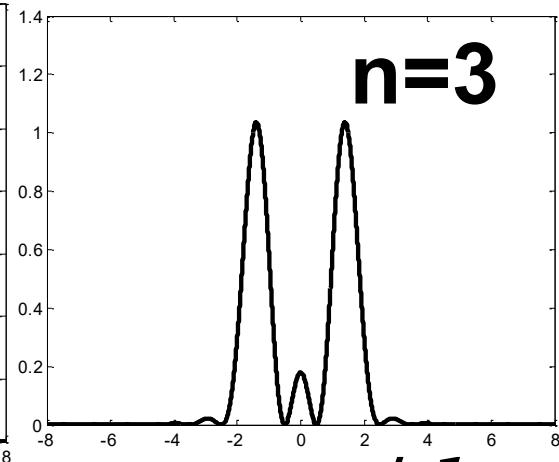
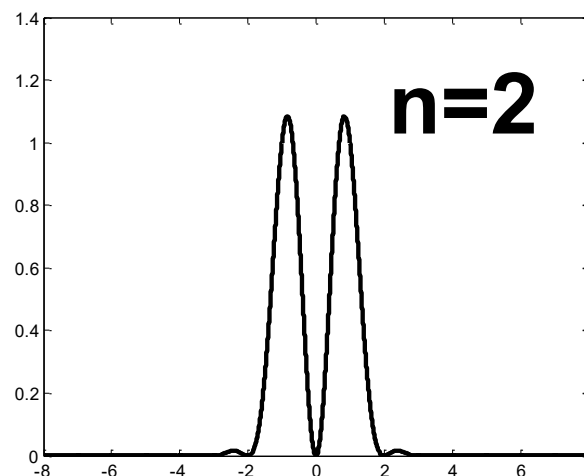
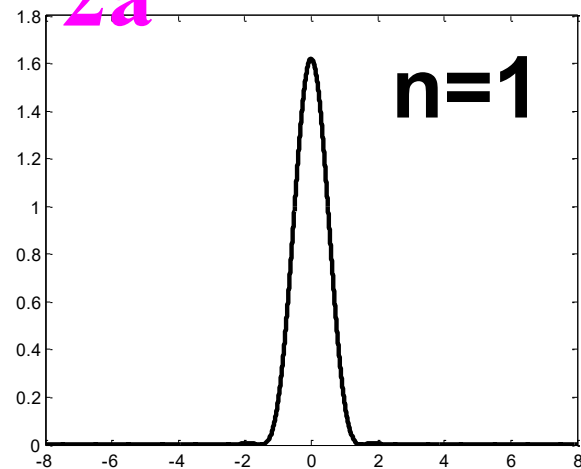
动量几率幅为 $c(p) = \int_{-\infty}^{\infty} \varphi_p^*(x) \psi_n(x) dx$

$$= \int_0^a \frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i}{\hbar}px} \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) dx$$

$$= \int_0^a \frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i}{\hbar}px} \sqrt{\frac{2}{a}} \frac{e^{i\frac{n\pi x}{a}} - e^{-i\frac{n\pi x}{a}}}{2i} dx$$

动量几率密度

$$|c_p|^2 \frac{h}{2a}$$

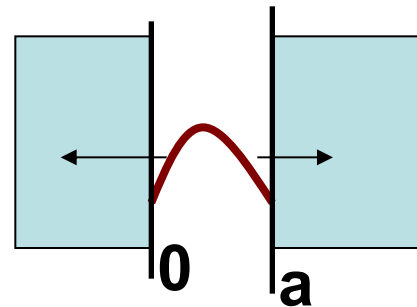


pa/h

pa/h

pa/h

例8. 一个粒子在 $t < 0$ 时刻被束缚在一个一维无限深势阱中，并处于基态。假设在 $t = 0$ 时刻势阱被突然关闭（在全空间势能变为零），求以后时刻 t ($t > 0$) 粒子的波函数及动量的几率密度。



分析：首先求解一维无限深势阱中的基态波函数（方法即前面学过的分离变量法）

$$\psi_1(x) = \begin{cases} \sqrt{\frac{2}{a}} \sin \frac{\pi}{a} x & (0 < x < a) \\ 0 & (\text{others}) \end{cases} \quad E_1 = \frac{\hbar^2}{2\mu} \frac{\pi^2}{a^2}$$

($t > 0$ 为自由粒子)

$$\psi(x, t) = \int_{-\infty}^{\infty} \underline{c(p)} \varphi_p(x) e^{-iEt/\hbar} dp \quad (t > 0)$$

$\psi_1(x)$


$(E = p^2 / 2\mu)$

时间上连续: $\psi(x, 0+) = \psi(x, 0-) = \psi_1(x)$

$$c(p) = \int_{-\infty}^{\infty} \varphi_p^*(x) \psi(x, 0+) dx = \int_{-\infty}^{\infty} \varphi_p^*(x) \psi_1(x) dx$$

$$\varphi_p(x) = \sqrt{\frac{1}{2\pi\hbar}} e^{ipx/\hbar}$$

$$\psi_1(x) = \begin{cases} \sqrt{\frac{2}{a}} \sin \frac{\pi}{a} x & (0 < x < a) \\ 0 & (\text{others}) \end{cases}$$

$$c(p) = \int_0^a \sqrt{\frac{2}{a}} \sin \frac{\pi}{a} x \sqrt{\frac{1}{2\pi\hbar}} e^{-ipx/\hbar} dx$$

$$= \frac{1}{\left(\frac{2pa}{\hbar}\right)^2 - 1} \left(1 + e^{-i2\pi\left(\frac{pa}{\hbar}\right)} \right)$$

$$w(p) = |c(p)e^{-iEt/\hbar}|^2 = |c(p)|^2$$

与势阱关闭前相同（时间上连续），

撤去势阱后动量几率密度不随t改变。

讨论：不变的原因是什么？（是因为定态还是守恒量？）

* 讨论:

动量是否一直是守恒量

(提示: 考虑势阱的作用, $[p, U(x)] \neq 0$)

能量呢? (势阱撤去前后比较)

(提示: H 含 t (系统与外界有能量交换))

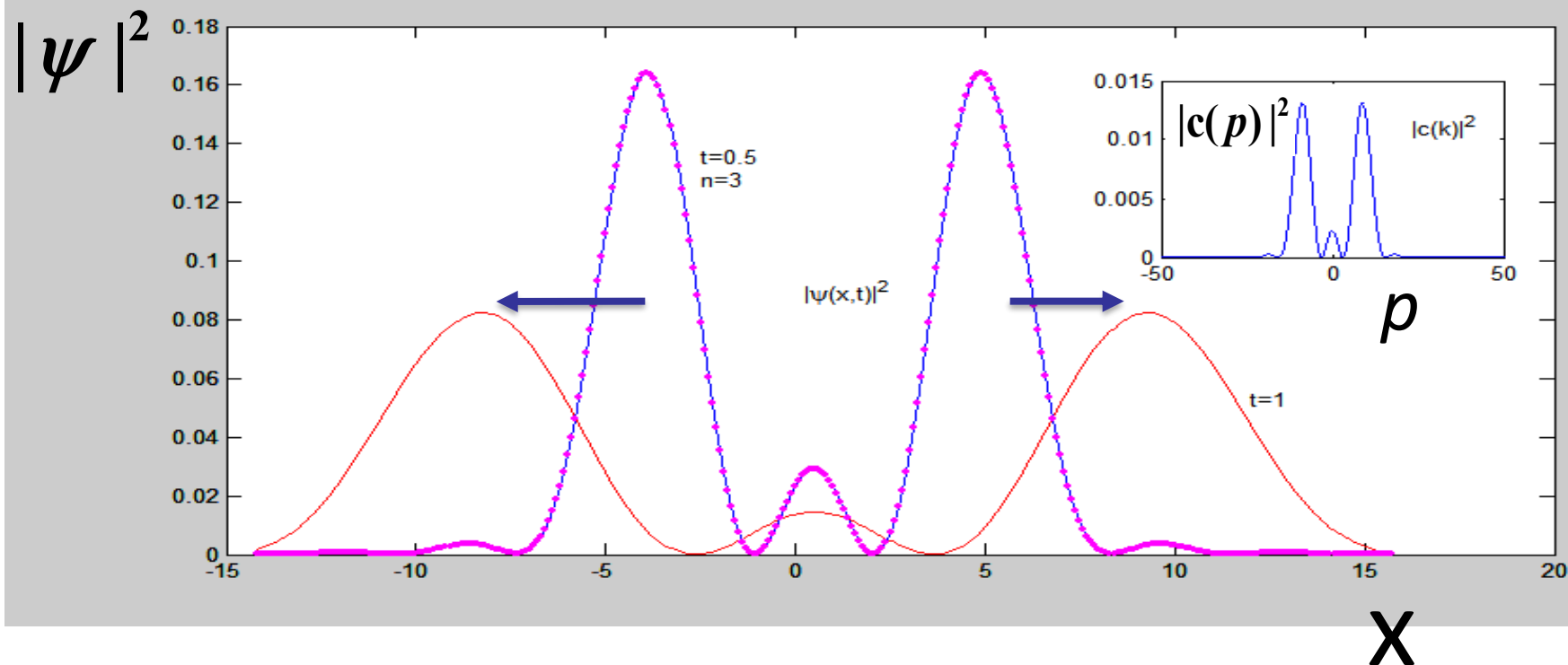
撤去前后都是定态吗?

(提示: H 变化, 无限深势阱的定态不是自由粒子的定态)

空间几率密度

无限深势阱 $n=3$ 撤去势阱后空间几率密度随 t 的演化。（图为数值计算，结果仅供参考参考）

$$\psi(x,t) = \int_{-\infty}^{\infty} c(p) \sqrt{\frac{1}{2\pi\hbar}} e^{ipx/\hbar - iEt/\hbar} dp \quad (t > 0) \quad (E = p^2 / 2\mu)$$



*附：作业补充题 4.6 解答

- 考虑一个放在恒定磁场中的电子，设磁感应强度为**B**，方向沿着**z**坐标轴正方向。分析电子自旋角动量的平均值是否随时间变化？如果变化，是怎样的运动？

提示：哈密顿量 $H = -\vec{M}_s \cdot \vec{B}$

$$\text{自旋磁矩: } \vec{M}_s = -\frac{e}{m_e} \hat{S}$$

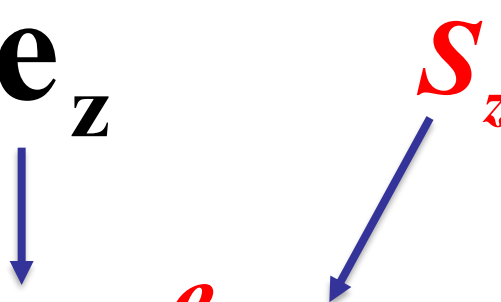
解：

一. 先求波函数

方法1 先求H本征态（定态问题），
然后写出通解（给出含时间的一般状态的波函数）

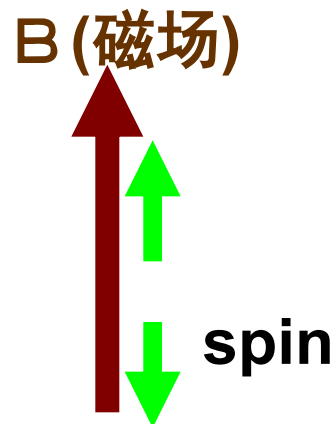
$$H = -\vec{M}_s \cdot \vec{B} = -\frac{-e}{m_e} \hat{\vec{S}} \cdot \vec{B} = \frac{e}{m_e} \hat{\vec{S}} \cdot \vec{B}$$

$$\mathbf{B} = B_0 \mathbf{e}_z$$

$$\mathbf{S}_z = \frac{\hbar}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$


$$H = \frac{e}{m_e} \hat{\mathbf{S}} \cdot \mathbf{B} = \frac{e}{m} S_z B_0$$

$$= \frac{eB_0}{m} \frac{\hbar}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



$$H \text{ 本征值: } E_\alpha = \frac{eB_0}{m} \frac{\hbar}{2}; \quad E_\beta = -\frac{eB_0}{m} \frac{\hbar}{2}$$

H本征矢

$$|\alpha\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{—————}$$

$$|\beta\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{—————}$$

$$|\psi(t)\rangle = c_\alpha e^{-\frac{i}{\hbar}E_\alpha t} |\alpha\rangle + c_\beta e^{-\frac{i}{\hbar}E_\beta t} |\beta\rangle = \begin{bmatrix} c_\alpha e^{-\frac{i}{\hbar}E_\alpha t} \\ c_\beta e^{-\frac{i}{\hbar}E_\beta t} \end{bmatrix}$$

方法2

$$|\Psi(t)\rangle = e^{-\frac{i}{\hbar}Ht} |\Psi(0)\rangle = e^{-\frac{i}{\hbar}Ht} \sum_{\mathbf{n}} c_{\mathbf{n}} |\psi_{\mathbf{n}}\rangle$$

$$= \sum_{\mathbf{n}} c_{\mathbf{n}} e^{-\frac{i}{\hbar}Ht} |\psi_{\mathbf{n}}\rangle = \sum_{\mathbf{n}} c_{\mathbf{n}} e^{-\frac{i}{\hbar}E_{\mathbf{n}}t} |\psi_{\mathbf{n}}\rangle$$

$$= c_{\alpha} e^{-\frac{i}{\hbar}E_{\alpha}t} |\alpha\rangle + c_{\beta} e^{-\frac{i}{\hbar}E_{\beta}t} |\beta\rangle = \begin{bmatrix} c_{\alpha} e^{-\frac{i}{\hbar}E_{\alpha}t} \\ c_{\beta} e^{-\frac{i}{\hbar}E_{\beta}t} \end{bmatrix}$$

方法3 直接求解含时间薛定谔方程

$$H = \frac{\textcolor{red}{e}}{\textcolor{red}{m}} \textcolor{red}{S}_z B_0 = \frac{eB_0}{m} \frac{\hbar}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} E_\alpha & 0 \\ 0 & E_\beta \end{bmatrix}$$

$$i\hbar \frac{\partial}{\partial t} \begin{bmatrix} c_\alpha(t) \\ c_\beta(t) \end{bmatrix} = \begin{bmatrix} E_\alpha & 0 \\ 0 & E_\beta \end{bmatrix} \begin{bmatrix} c_\alpha(t) \\ c_\beta(t) \end{bmatrix} \quad \begin{aligned} E_\alpha &= \frac{eB_0}{m} \frac{\hbar}{2} \\ E_\beta &= -\frac{eB_0}{m} \frac{\hbar}{2} \end{aligned}$$

$$i\hbar \dot{c}_\alpha = E_\alpha c_\alpha$$

$$c_\alpha(t) = c_\alpha(0) e^{-\frac{i}{\hbar} E_\alpha t},$$

$$i\hbar \dot{c}_\beta(t) = E_\beta c_\beta \longrightarrow$$

$$c_\beta(t) = c_\beta(0) e^{-\frac{i}{\hbar} E_\beta t}$$

二. 求自旋三个分量的平均值

$$\overline{S_z} = (c_\alpha^*, c_\beta^*) \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} c_\alpha \\ c_\beta \end{pmatrix} = \frac{\hbar}{2} (|c_\alpha|^2 - |c_\beta|^2)$$

$$\overline{S_x} = (c_\alpha^*, c_\beta^*) \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} c_\alpha \\ c_\beta \end{pmatrix} = \frac{\hbar}{2} (c_\alpha^* c_\beta + c_\beta^* c_\alpha)$$

$$\overline{S_y} = (c_\alpha^*, c_\beta^*) \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} c_\alpha \\ c_\beta \end{pmatrix} = \frac{\hbar}{2i} (c_\alpha^* c_\beta - c_\beta^* c_\alpha)$$

$$c_\alpha(t) = c_\alpha(0) e^{-\frac{i}{\hbar} E_\alpha t}, \quad c_\beta(t) = c_\beta(0) e^{-\frac{i}{\hbar} E_\beta t}, \quad \text{代入上式}$$

$$\text{并设 } c_\alpha(0)^* c_\beta(0) = |c_\alpha(0)| |c_\beta(0)| \exp(i\delta)$$

$c_\alpha(0)$ 、 $c_\beta(0)$ 两个几率幅的初始相位差

$$\langle S_x \rangle = \frac{\hbar}{2} 2 |c_\alpha(0)| |c_\beta(0)| \cos(\omega_0 t + \delta)$$

$\langle \dots \rangle$ 代表平均值

$$\langle S_y \rangle = \frac{\hbar}{2} 2 |c_\alpha(0)| |c_\beta(0)| \sin(\omega_0 t + \delta)$$

$$\langle S_z \rangle = \frac{\hbar}{2} (|c_\alpha(0)|^2 - |c_\beta(0)|^2);$$

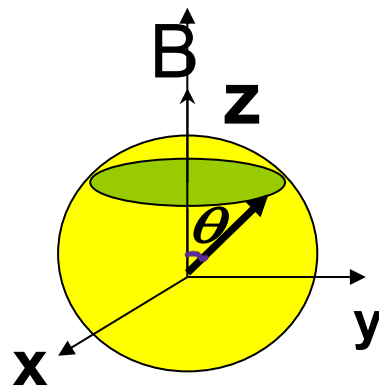
(进动角频率)

(S_z 是守恒量, 也可由 $[S_z, H=0]$ 得知)

$$E_\alpha = \frac{eB_0}{m} \frac{\hbar}{2} = \frac{\hbar}{2} \omega_0$$

$$E_\beta = -\frac{eB_0}{m} \frac{\hbar}{2} = -\frac{\hbar}{2} \omega_0$$

$$\omega_0 = \frac{E_\alpha - E_\beta}{\hbar} = \frac{eB_0}{m}$$



最后答案

进动周期: $T = \frac{2\pi}{\omega_0}$

$$\langle S_x \rangle = \frac{\hbar}{2} 2 |c_\alpha(0)| |c_\beta(0)| \cos(\omega_0 t + \delta) \text{ (周期运动)}$$

$$\langle S_y \rangle = \frac{\hbar}{2} 2 |c_\alpha(0)| |c_\beta(0)| \sin(\omega_0 t + \delta) \text{ (周期运动)}$$

$$\langle S_z \rangle = \frac{\hbar}{2} (|c_\alpha(0)|^2 - |c_\beta(0)|^2) \quad \text{(保持恒定)}$$

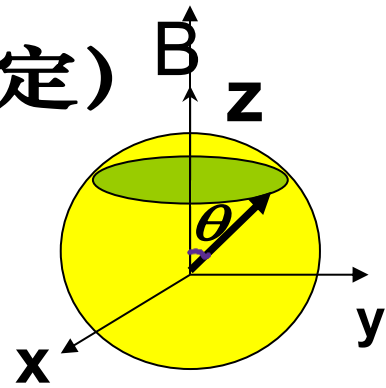
自旋平均值大小



$$|\langle \vec{S} \rangle| = \sqrt{\langle S_x \rangle^2 + \langle S_y \rangle^2 + \langle S_z \rangle^2}$$

$$= \frac{\hbar}{2} \sqrt{(|c_\alpha(0)|^2 - |c_\beta(0)|^2)^2 + 4 |c_\alpha(0)|^2 |c_\beta(0)|^2}$$

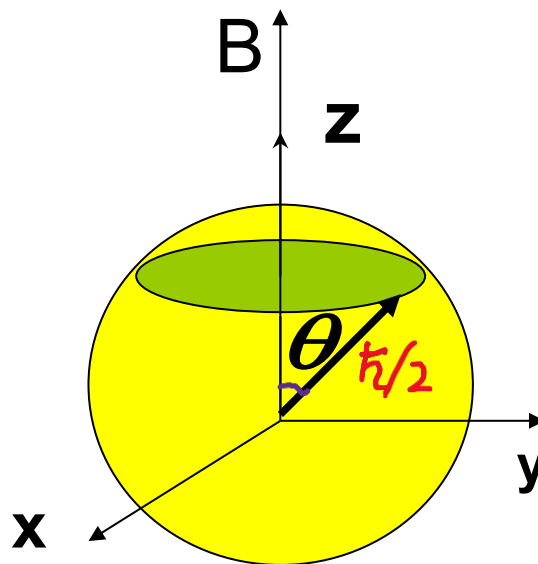
$$= \frac{\hbar}{2} \sqrt{(|c_\alpha(0)|^2 + |c_\beta(0)|^2)^2} = \frac{\hbar}{2}$$



自旋平均值方向

$$|\langle \vec{S} \rangle| = \frac{\hbar}{2} \text{ (保持恒定)}$$

$$\cos \theta = \frac{\langle S_z \rangle}{\langle \vec{S} \rangle} = |c_\alpha(0)|^2 - |c_\beta(0)|^2 \text{ (保持恒定)}$$



$\langle \vec{S} \rangle$ 大小恒定 ($= \frac{\hbar}{2}$)，围绕磁场 B 旋转，

进动周期 $\frac{2\pi}{\omega_0}$ ，和磁场方向的夹角 θ 不变。

波函数

$$|\Psi(t)\rangle = \begin{bmatrix} c_\alpha e^{-\frac{i}{\hbar} E_\alpha t} \\ c_\beta e^{-\frac{i}{\hbar} E_\beta t} \end{bmatrix} = \begin{bmatrix} c_\alpha e^{-\frac{i}{2} \omega_0 t} \\ c_\beta e^{\frac{i}{2} \omega_0 t} \end{bmatrix}$$

自旋角动量平均值 $\langle \vec{S} \rangle$ 随

时间变化的周期是 $T = \frac{2\pi}{\omega_0}$,

但 $|\Psi(t+T)\rangle = -|\Psi(t)\rangle$;

$|\Psi(t+2T)\rangle = |\Psi(t)\rangle$ 。

(“转两圈才复原!”)

$$\begin{array}{l} E_\alpha = \frac{\hbar}{2} \omega_0 \text{ —————} \\ 0 \text{ - - - - -} \\ E_\beta = -\frac{\hbar}{2} \omega_0 \text{ —————} \end{array}$$